

Project Overview: Model Selection and Regularization Techniques

1. Comparing Model Selection Techniques

This section explores three model selection strategies: best subset selection, forward stepwise selection, and backward stepwise selection.

Key Findings:

- **Training RSS:** Best subset selection gives the lowest training RSS for a model with k predictors, as it considers all possible combinations.
- **Test RSS:** No single method consistently minimizes test RSS; overfitting (especially in best subset) can increase test error. Validation techniques like cross-validation are recommended.
- **Predictor Nesting Relationships:**
 - Forward and backward selection yield nested models within their own sequences.
 - Forward and backward models are not guaranteed to be subsets of each other.
 - Best subset models with k predictors are always subsets of their $k+1$ models.

2. Bias-Variance Tradeoff and Regularization (Lasso)

Exploring how the regularization parameter $\mathbf{s}$ (in Lasso or constrained regression) affects model behavior:

Metric	Behavior as $\mathbf{s}$ increases
Training RSS	Steadily decreases (model fits better)
Test RSS	Decreases then increases (U-shape)
Variance	Steadily increases
Squared Bias	Steadily decreases
Irreducible Error	Remains constant

Conclusion: Increasing model flexibility reduces bias but increases variance, emphasizing the need for tuning via cross-validation.

3. Polynomial Feature Selection (Simulated Example)

Goal: Identify the best model from a simulated dataset where the true relationship is:

$$Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \beta_3 X^3 + \epsilon$$

Approach:

- Generated polynomial features up to X^{10}
- Applied:
 - **Best Subset Selection**
 - **Forward Stepwise Selection**
 - **Backward Stepwise Selection**
- Evaluated models using **Cp**, **BIC**, and **Adjusted R^2**

Results:

- All methods identified the correct 3-predictor model (X, X^2, X^3)
- Cp and BIC selected model with 3 predictors
- Adjusted R^2 peaked at 3 predictors
- The estimated coefficients closely matched the true values

4. Lasso Regularization and Variable Selection

Goal:

Use Lasso to select variables and estimate coefficients from the same polynomial dataset.

Method:

- Fitted Lasso using `cv.glmnet()` to select optimal λ
- Model matrix included all polynomial features up to X^{10}

Findings:

- Lasso identified X, X^2 , and X^3 with coefficients close to the true values
- Unimportant predictors had coefficients shrunk to zero
- Lasso demonstrated its strength in sparse model identification

5. Alternative True Model: $Y = \beta_0 + \beta_7 X^7 + \epsilon$

Tested Using:

- Best Subset Selection
- Lasso with Cross-Validation

Key Observations:

- Both methods successfully identified X^7 as the true predictor
- Lasso excluded irrelevant features by shrinking their coefficients to zero
- Confirms Lasso's robustness in identifying dominant variables in nonlinear relationships

6. Best Subset Selection on Simulated High-Dimensional Data

Setup:

- Generated data with 20 predictors and true coefficients for only 3 variables
- Split into training (100 samples) and testing sets

Results:

- Training MSE: Decreased steadily with model size
- Test MSE: U-shaped; optimal at intermediate model size
- Optimal Model Size: 1 (based on test MSE)
- Coefficient Recovery: Closest alignment with true model occurs at optimal test MSE size

7. Evaluating Coefficient Accuracy Across Models

Metric Used:

Sum of squared differences between true and estimated coefficients across model sizes

Insights:

- Plot showed a U-shaped curve, mimicking test MSE trend
- Underfitting (small models): Large coefficient error due to missing true predictors
- Overfitting (large models): Larger error due to inclusion of irrelevant variables
- Minimum error occurred at the model size that also minimized test MSE